

Tristan Hoellinger

JPL Postdoctoral Fellow · Cosmology · NASA/Caltech

Positions

2026–2030 **Postdoctoral Fellow in Cosmology**, **Jet Propulsion Laboratory**, NASA/Caltech
Expertise: Field-level Inference, Observational Cosmology. PIs: Olivier Doré and Dida Markovic.

Collaborations

Since 2023 **Euclid Consortium**, *active role in ILI-GC-DR1 (PI: D. Bartlett)*.
Since 2023 **Aquila Consortium**, *monthly telecons, bi-annual one-week workshops*.

Leadership

2024, 2025 **Local Organizing Committee (LOC)**, **Aquila Workshops**, IAP, Paris
2023–2024 **Organizer**, *weekly meetings with invited colloquium speakers*, IAP, Paris
2024 **Outreach workshop at the Fête de la Science 2024**, IAP, Paris
2023 **Member of the Paris LOC**, **ML-IAP/CCA conference**, Hybrid IAP/Flatiron, New York
2023–2025 **Teaching Mathematics**, *140 hours on-site*, **Third year BSc Mathematics**, Sorbonne University

Grants & Awards

2023 Sorbonne "Politique Scientifique" PhD Fellowship; sole recipient in Astrophysics (~€100k).
2016–2020 Undergraduate scholarships: merit-based & social criteria (~€7k/year).

Education

2023–2026 **PhD in Astronomy & Astrophysics**, **Institut d'Astrophysique de Paris**, Sorbonne Université
Advisors: Florent Leclercq and Guilhem Lavaux.
2016–2022 **BSc/MSc in Applied Mathematics & MSc in Hybrid AI**, **N7 & INSA Toulouse**, France
2019–2020 **Academic Exchange in Fundamental Physics**, Paris-Saclay University

Open-Source Software

SelfiSys **Principal developer** and maintainer of SelfiSys, a Python package for diagnosing systematic effects in forward models of galaxy surveys, with **full documentation** and **examples**.
Simbelmynë A hierarchical probabilistic simulator to generate synthetic galaxy survey data. I **lead the development of a fast P3M gravity module**, to be open-sourced in a next release. Principal developer: F. Leclercq.

Publications

★ 2025 **Diagnosing systematic effects using the inferred initial power spectrum**,
Hoellinger and Leclercq, *A&A* **699**, A224 (2025); [arXiv:2412.04443](https://arxiv.org/abs/2412.04443).
★ 2025 **A vorticity confinement correction for DG schemes applied to fluid flow problems**,
Hoellinger, Manueco, and Chapelier, *IJNMHFF* (2025), [10.1108/hff-11-2024-0854](https://doi.org/10.1108/hff-11-2024-0854), [\[open-access\]](#).
★ 2023 **Enhancer/gene relationships: Need for more reliable genome-wide reference sets**,
Hoellinger et al., *Frontiers in Bioinformatics* **3** (2023), [10.3389/fbinf.2023.1092853](https://doi.org/10.3389/fbinf.2023.1092853).
2020 **Data-Driven Simulation for Augmented Surgery**, Mendizabal et al.,
Co-author *Advanced Structured Materials* **132** (Springer), [10.1007/978-3-030-50464-9_5](https://doi.org/10.1007/978-3-030-50464-9_5).

Talks (Selected)

- Invited 2026 **Invited Seminar**, *SCAI – The Sorbonne Cluster for AI*, Paris, France
Trustworthy AI for Cosmology: Auditable Inference & Hybrid Solvers, 45 min.
- Invited 2025 **Invited Seminar**, *Observatoire Astronomique*, Strasbourg, France
Implicit Likelihood Cosmological Inference: Simulations & Diagnostics, 45 min.
- 2025 **Les Houches on Dark Universe**, *Les Houches School of Physics*, Les Houches, France
Accurate Small-Scale Dynamics in COLA, 20 min.
- Invited 2025 **Cosmology & Gravitation Seminar**, *Oskar Klein Centre (OKC)*, Stockholm, Sweden
Addressing Systematic Effects in Implicit Cosmological Inference, 30 min.
- Invited 2025 **Cosmology Group Meeting**, *Institute of Cosmos Science (ICC)*, Barcelona, Catalonia, Spain
Diagnosing systematic effects in field-based, implicit likelihood cosmological inference, 40 min.
- 2024 **X Meeting on Fundamental Cosmology**, *Universidad de Sevilla*, Seville, Spain
Lightening black-box models in field-based implicit likelihood cosmological inference, 15 min.

Conferences, Schools and Workshops

- 2023–2026 **Aquila Consortium Workshops**, twice yearly, Paris, Oxford, Stockholm
- 2023, 2024 **Rodolphe Clédassou School**, Ronces-les-Bains & Hendaye, France
- 2024 **MaxEnt 2024**, *Faculty of Economics*, Ghent University, Ghent, Belgium
- 2023 **Future Cosmology**, *Institut d'Études Scientifiques de Cargèse*, Corsica

Interdisciplinary Research Projects

- 2022–2023 **Master's Thesis**, *ONERA - The French Aerospace Lab*, Châtillon
DG schemes for massively parallel direct simulation of turbulent fluid dynamics. Advisor: J.-B. Chapelier.
- 2020–2022 **Junior Bioinformatics Scientist**, *Inserm*, Toulouse
Statistical methods to map enhancers to their target genes in human genomes. Advisor: S. Djebali.
- 2021 **Visiting**, *Centre for Genomic Regulation*, Barcelona
A pipeline to find genetic variants linked to intron retention in blood cells. Advisor: D. Garrido.
- 2020 **Research Intern**, *Toulouse Mathematics Institute*, Toulouse
Stochastic calculus, quantum nondemolition measurements. Advisors: T. Benoist and C. Pellegrini.
- 2020 **Research Intern**, *IRAP*, Toulouse, Theoretical study of kinetic scale plasma turbulence. Advisor: P. Louarn.
- 2019 **Research Intern**, *Inria*, Strasbourg, Deep learning for data-driven surgery. Advisor: A. Mendizabal

References *Details upon request.*

J.-B. Chapelier, Researcher, ONERA.
F. Leclercq, Research Scientist, IAP.
G. Lavaux, Research Director, IAP.